

Problem Statement -

Problem Statement : Nutrition Agent

The Challenge

Good nutrition is the foundation of health, but individuals often struggle with **diet planning, portion control, and tracking nutritional intake**. Information about healthy diets, disease-specific nutrition, and food composition is scattered across unreliable sources. Without **personalized, real-time guidance**, many fail to maintain balanced diets or make informed food choices.

The Objective

Build an **AI-powered Nutrition Agent** that delivers **personalized dietary insights and guidance** for individuals, families, and health professionals. The solution should:

- **Personalized Diet Plans** – Recommend meals based on age, health conditions, allergies, cultural preferences, and fitness goals.
- **Food Data RAG Retrieval** – Fetch and summarize nutritional values (calories, macros, vitamins) from reliable sources.
- **Diet Tracking & Feedback** – Allow users to log meals via text, images, or voice, and provide instant nutritional analysis.
- **Preventive Health Focus** – Suggest diets for chronic disease management (e.g., diabetes-friendly or heart-healthy plans).
- **Multi-Agent System** –
 - **Nutrition Knowledge Agent** (fetches nutritional data)
 - **Diet Recommendation Agent** (personalized meal plans)
 - **Health Advisory Agent** (preventive health suggestions)
 - **Food Log & Feedback Agent** (meal tracking + analysis)
- **Visualization Dashboard** – Show daily/weekly nutrient intake, deficiencies, and progress toward health goals.